DeepSkin

INFO9023 - Spring 2025

Created By

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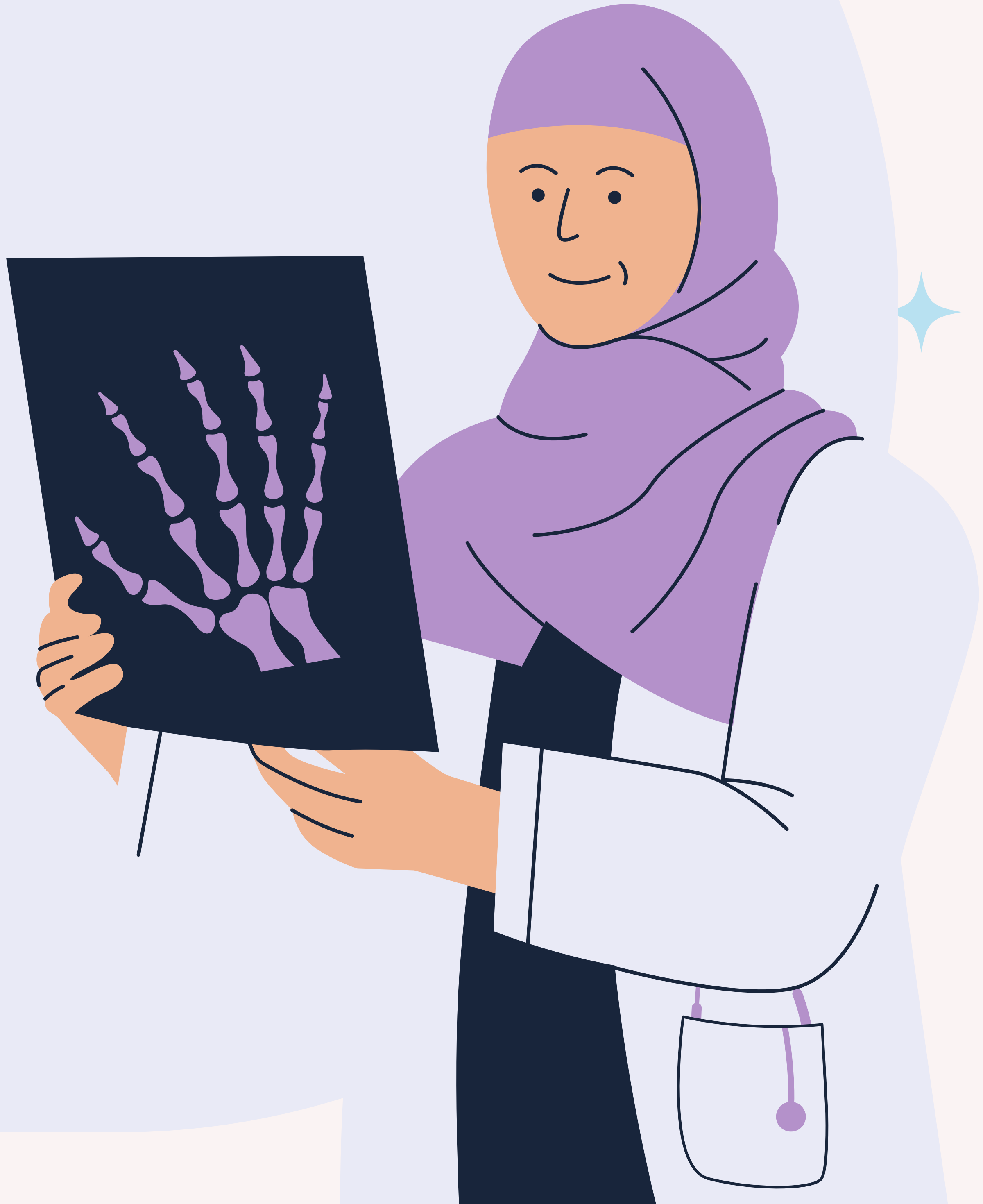
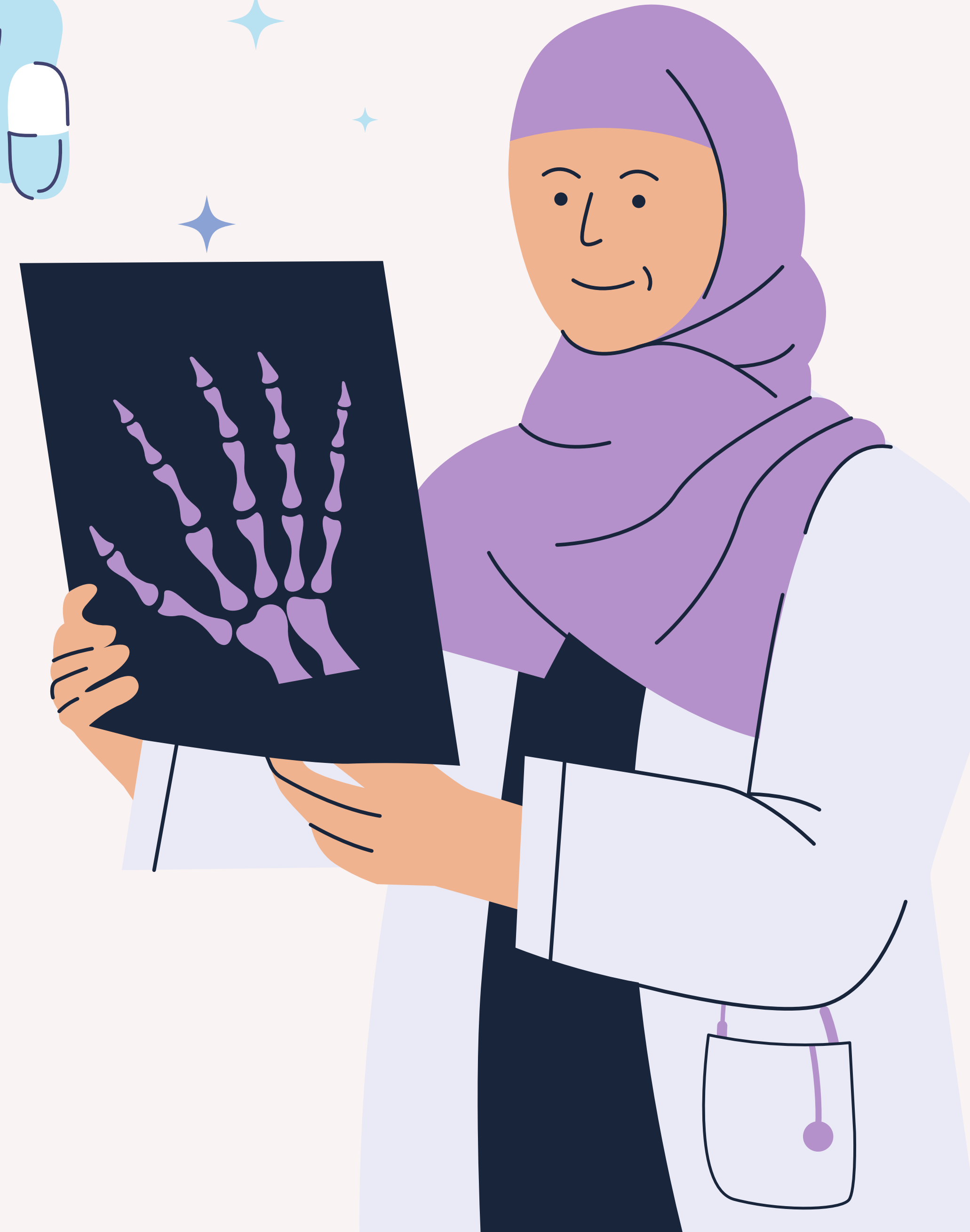
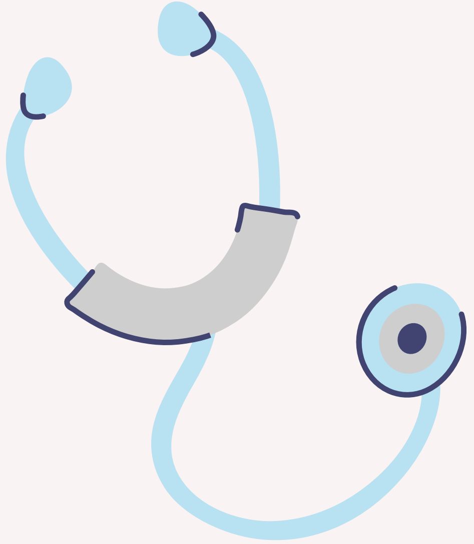


What is Deepskin ?



Current State of the medical world



Nowadays, obtaining a hospital appointment can take several weeks .

However, some cancers such as skin cancer which is one of the most common cancer in the world require **immediate** medical attention, and it is unreasonable to expect medical staff to handle such demand.

To deal with this problem, we proposed **DeepSkin** an ML-based solution designed to reduce the burden on healthcare providers by filtering **non-urgent** skin conditions from those that require immediate attention.

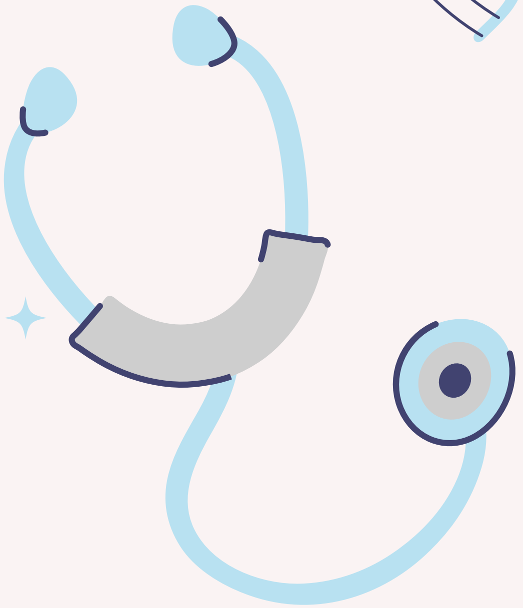
DataSet

The HAM10000 DataSet

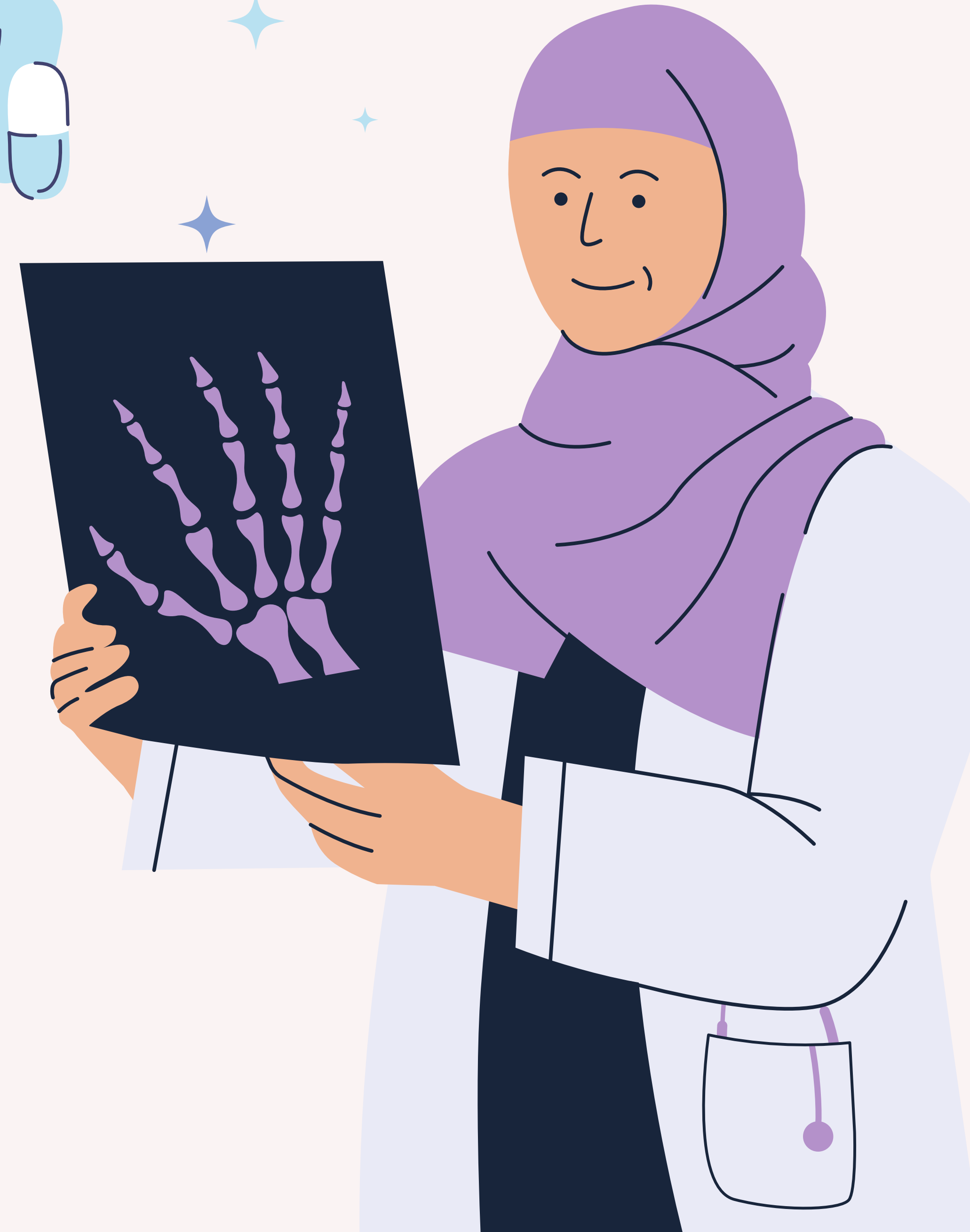
This dataset is a large collection of multi-source dermatoscopic images of pigmented lesions composed of 3 files

- The first one is a **metadata** file with the localization of the image, the type of the lesion as well as personal data such as the **age** / **sex** / **localization** of the lesion

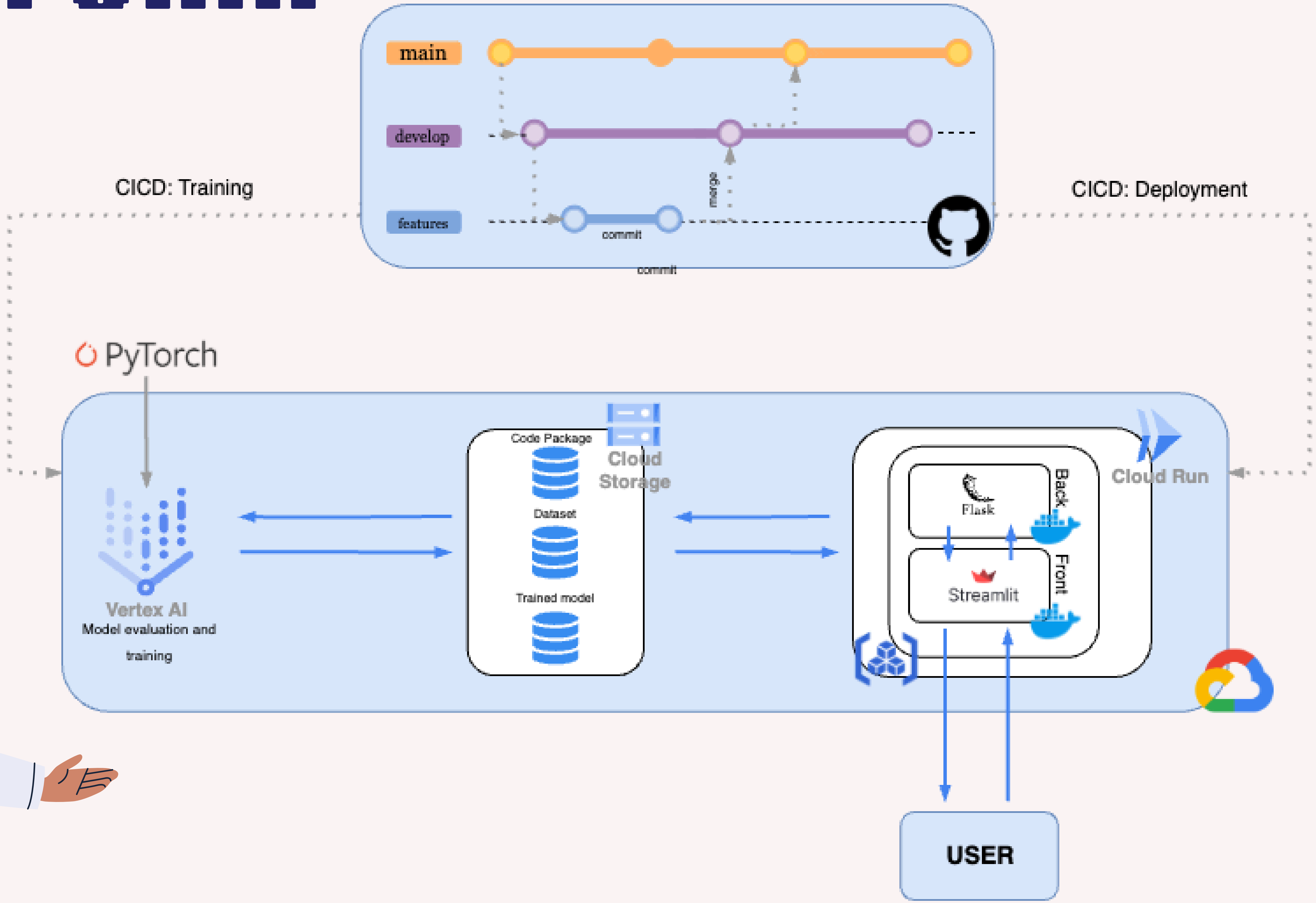
- The 2 other file are the image folders each of them containing 5000 skin lesions pictures



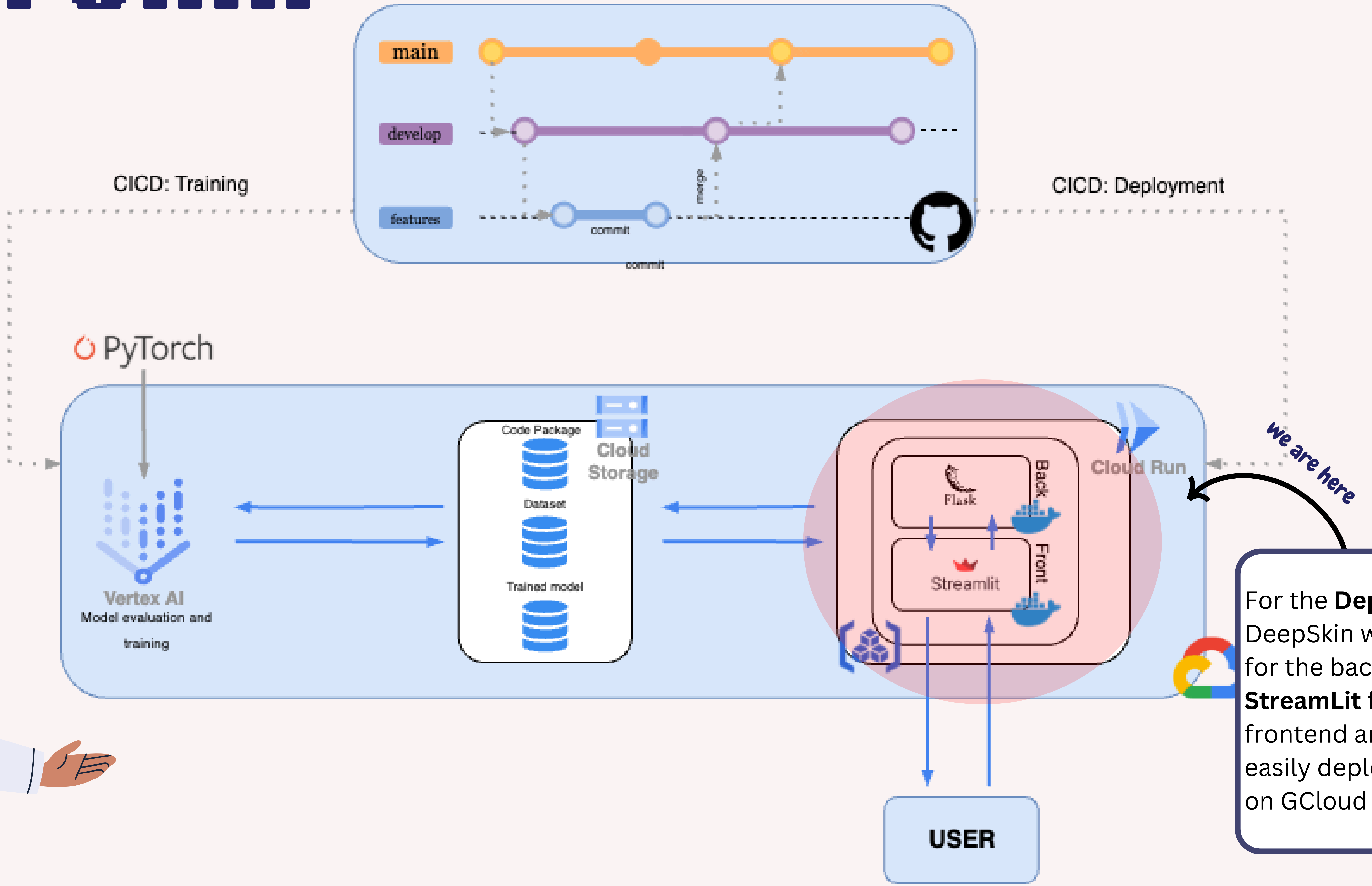

Global Overview

DEEPSKIN


main
develop
features
merge
commit
commit
CICD: Training
CICD: Deployment
PyTorch
Vertex AI
Model evaluation and
training
Code Package
Cloud
Storage
Dataset
Trained model
Back
Flask
Front
Streamlit
Cloud Run
USER

DEEPSKIN



For the **Deployment** of DeepSkin we used **Flask** for the back-end. **StreamLit** for the frontend and **Docker** to easily deploy everything on GCloud

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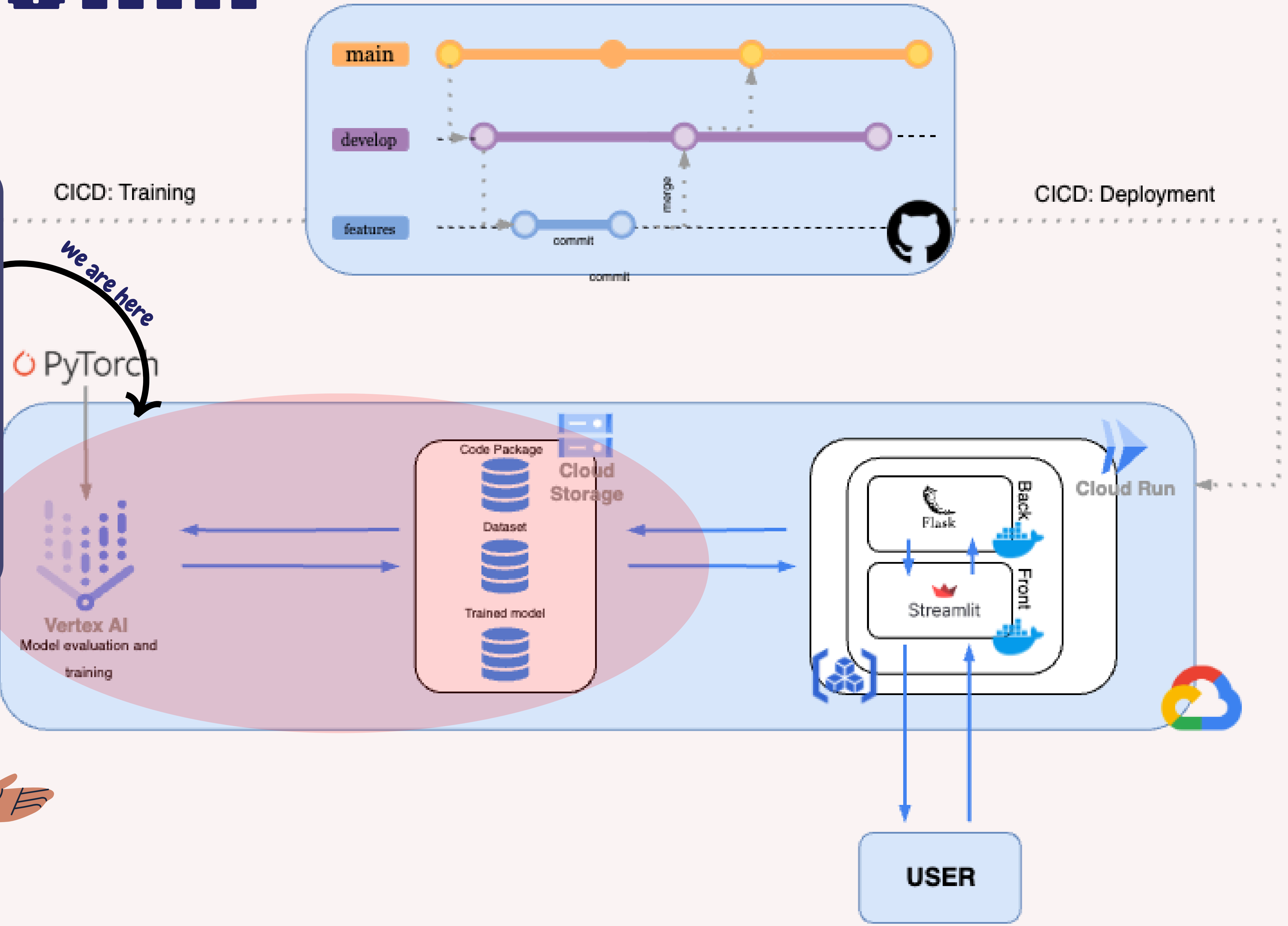
Regarding the **Training** part of our application we used **Vertex IA**.

Vertex requires to configure a job, execute our codebase with our dataset and return our trained model.

All of this can be done through **Cloud Storage** and Buckets



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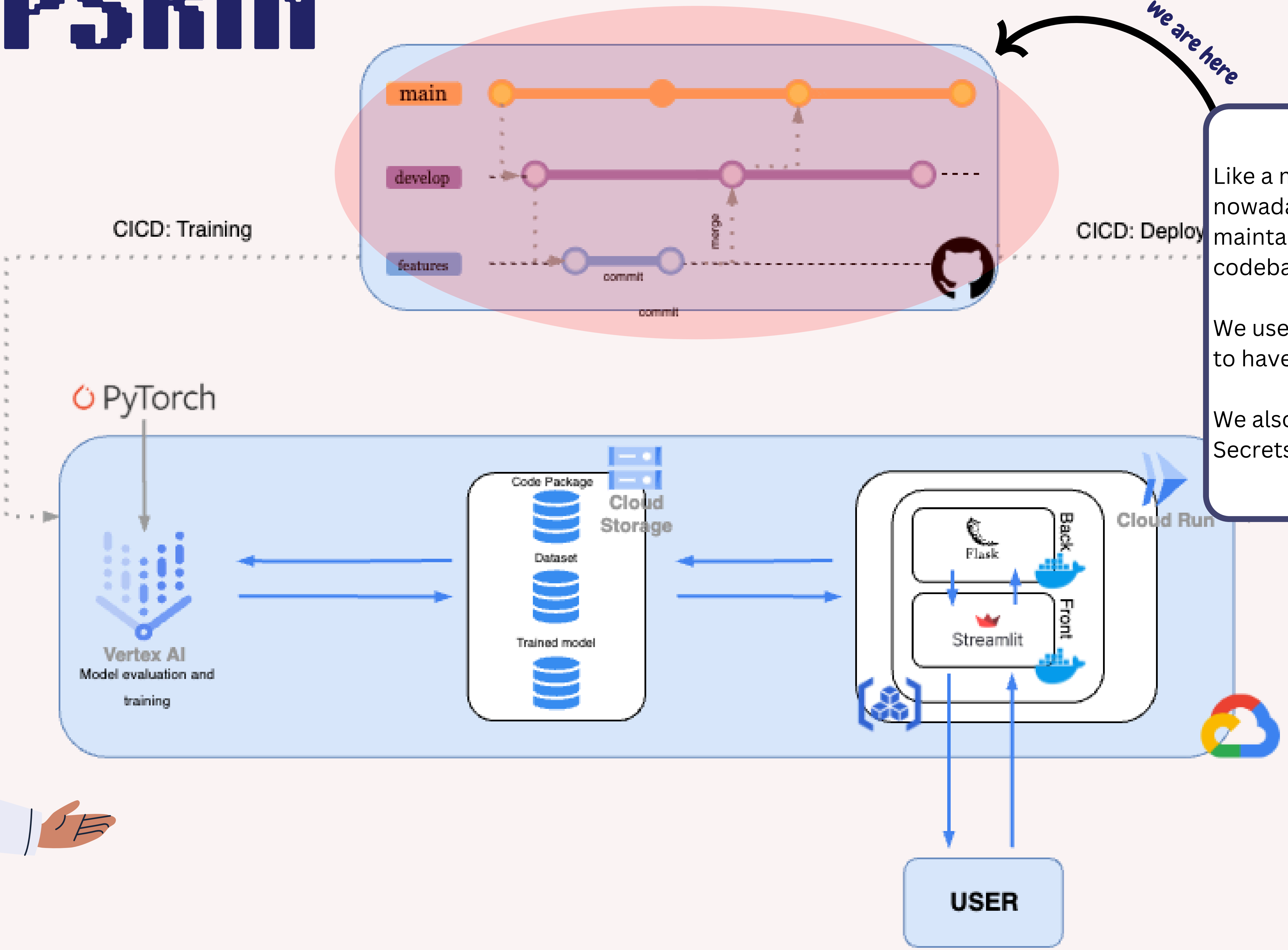
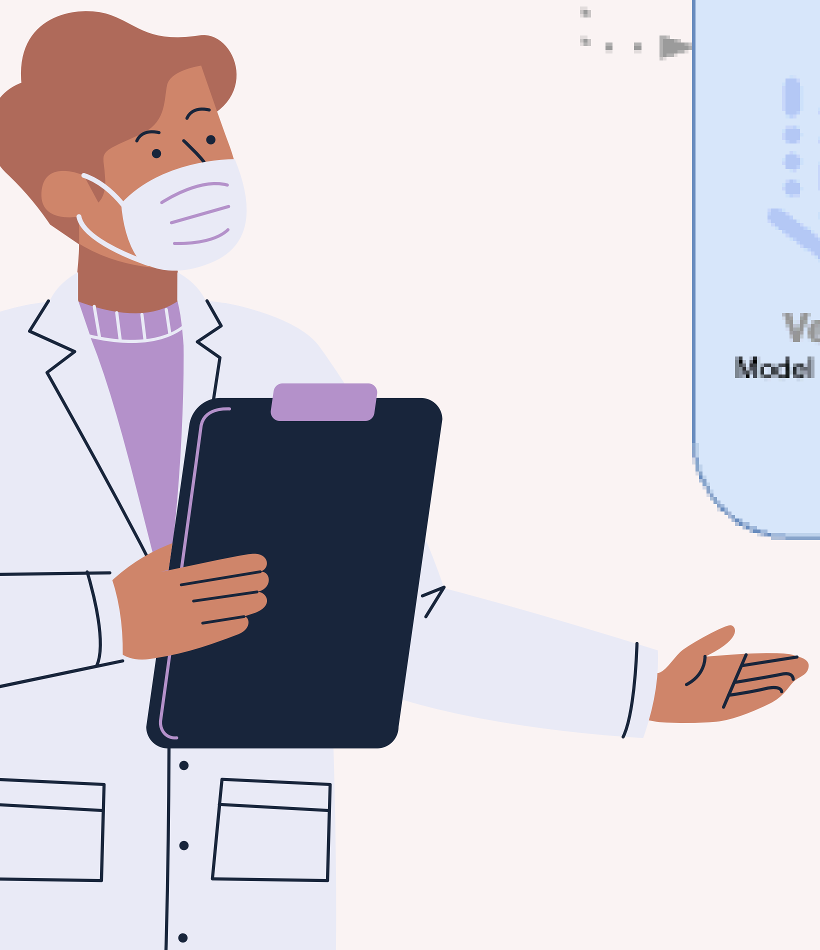


we are here

Like a majority of people nowadays we used Git to maintain and extend our codebase.

We used multiple branches to have a nice workflow.

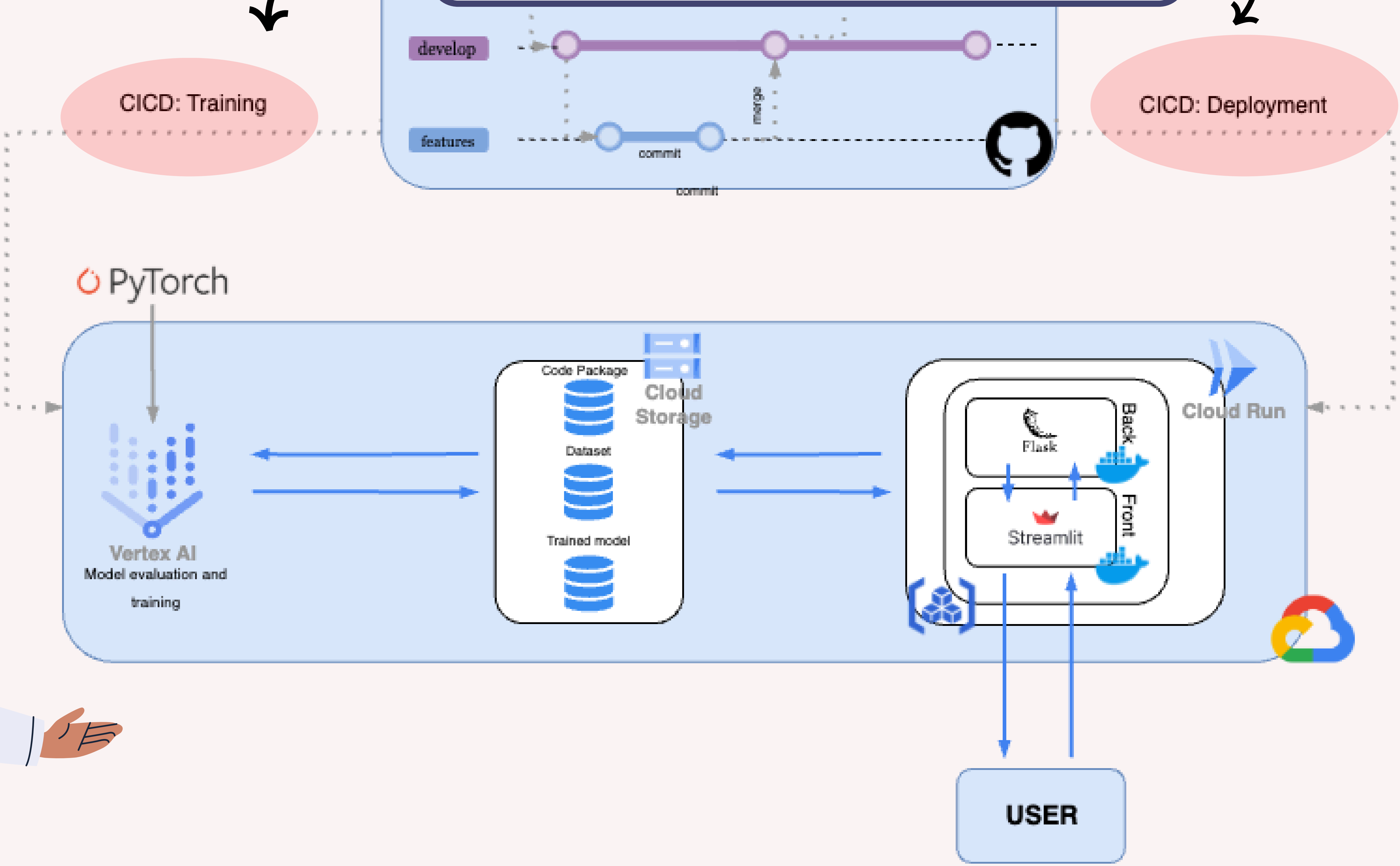
We also use the Github Secrets And Actions

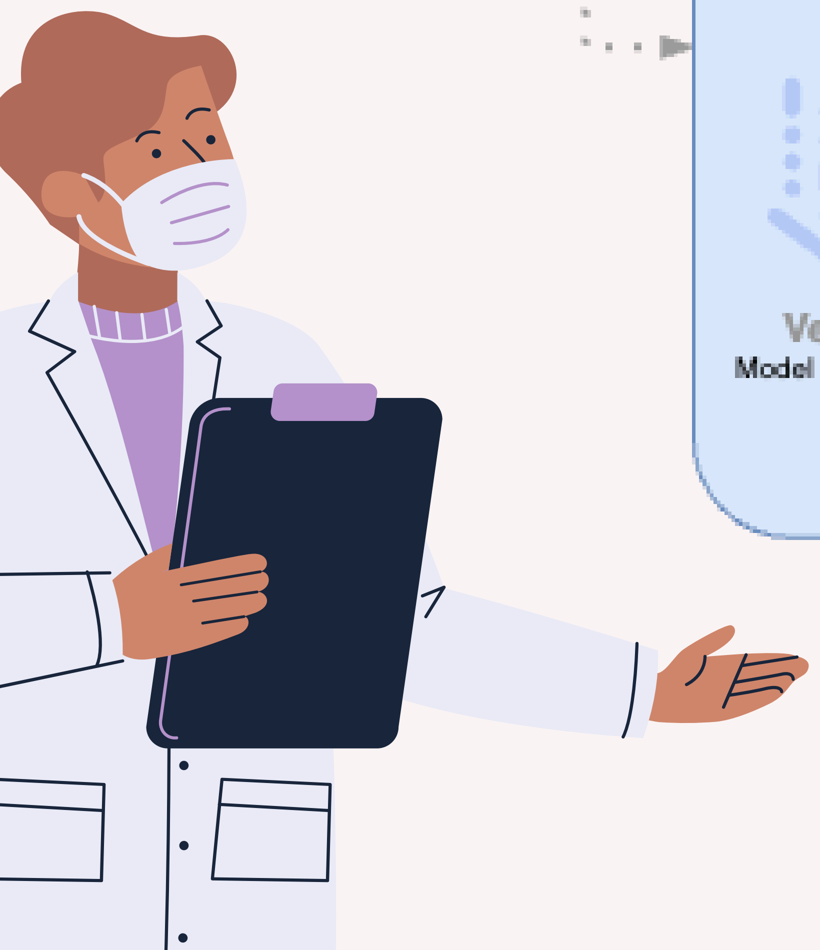


DEEPSKIN

For the CICD pipeline, we implement automatic training of our model, automatic deployments of both the front and the back.

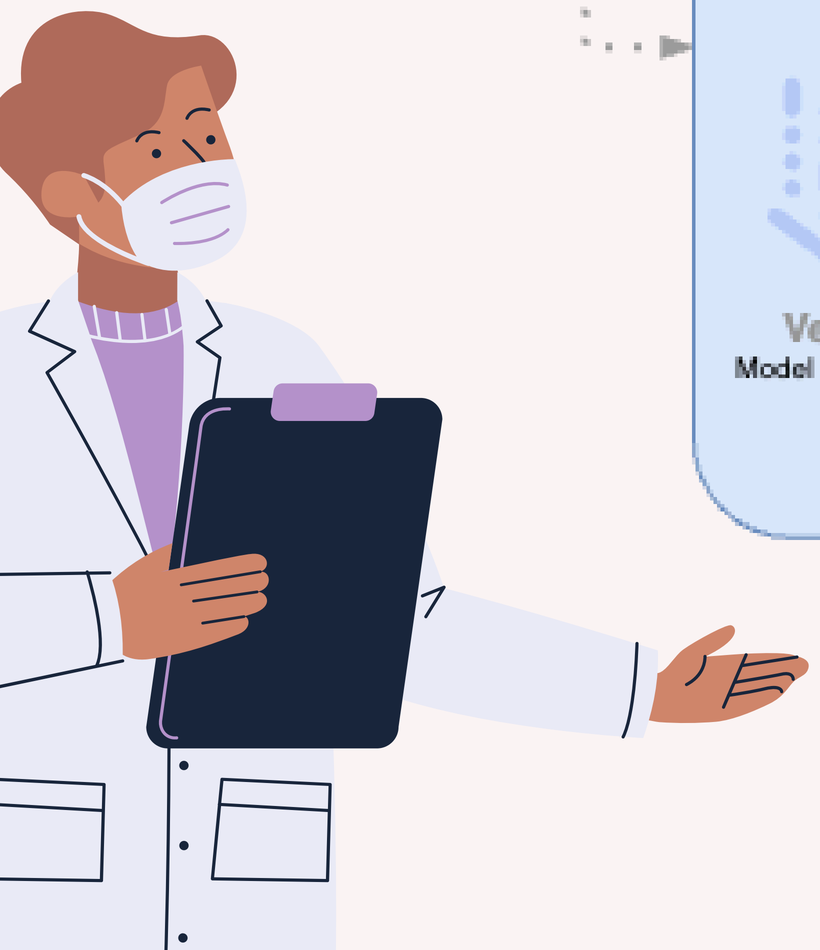
We used Pytest to test our API and ruff to format our code and we also use a pre-commit hook to ensure that bad quality code was never push to the repository.



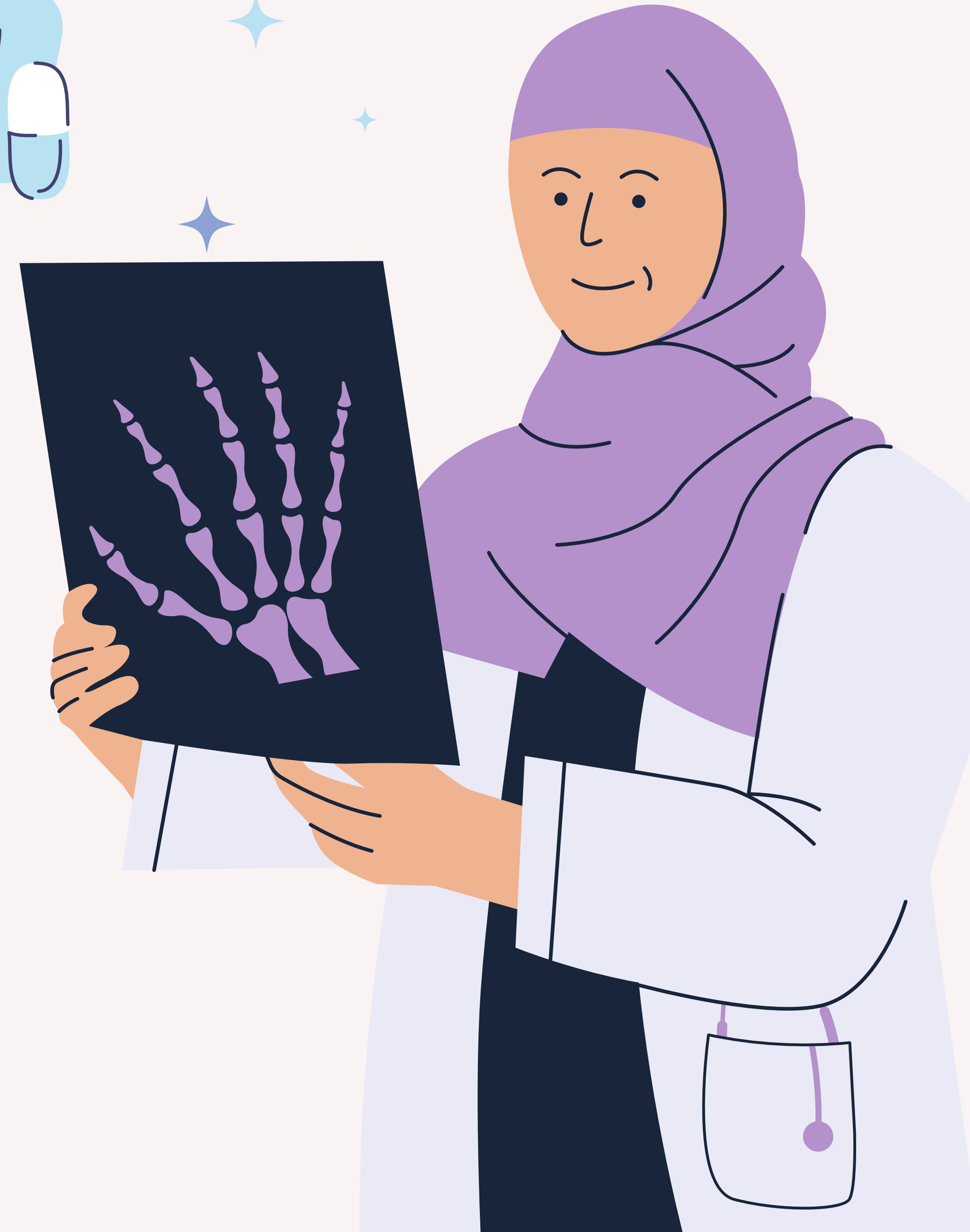


DEEPSKIN

DEMONSTRATION

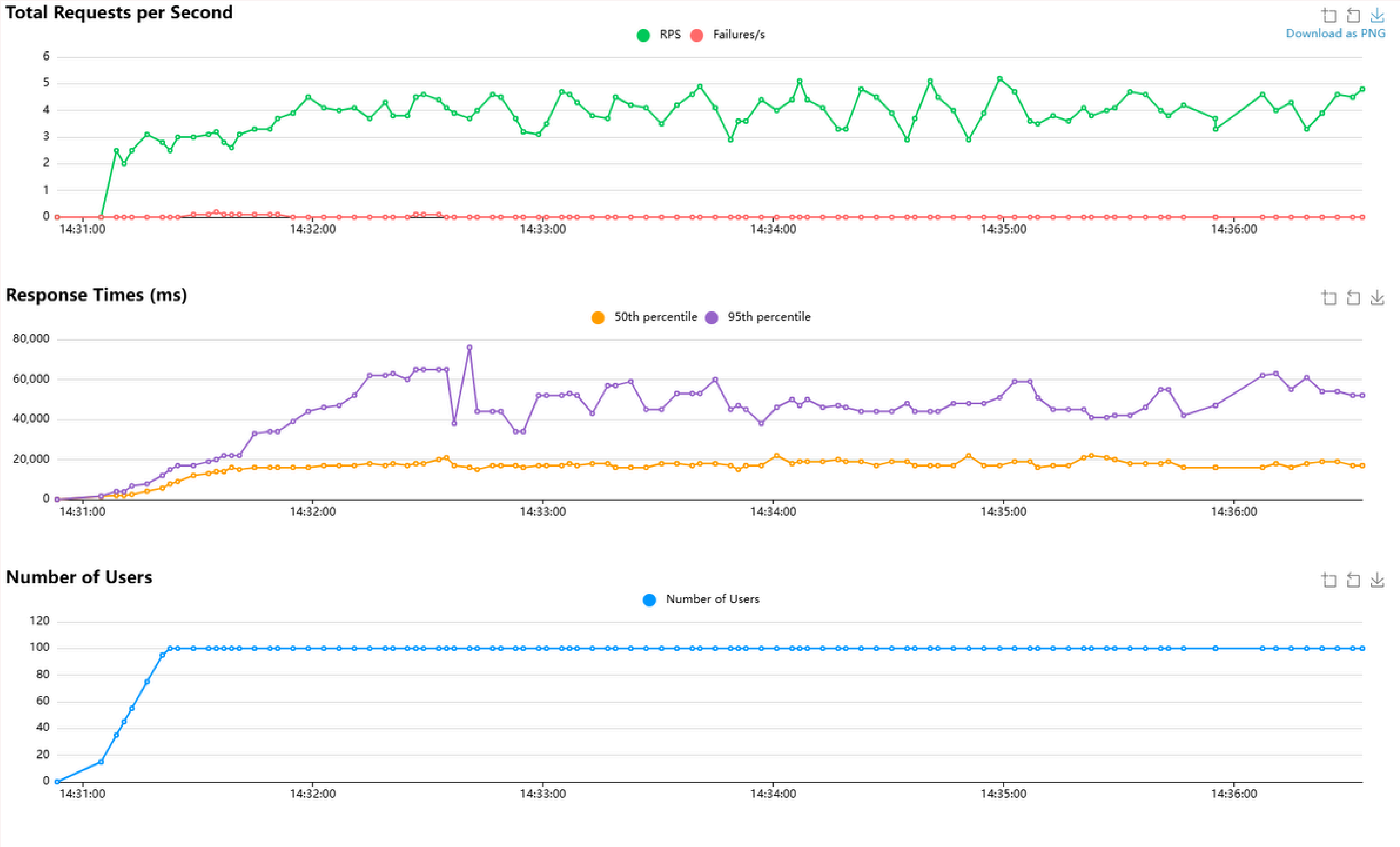


New Tools



DataSet

Locust is a load-testing tool that simulates virtual users and measures the performance of an API or web application.

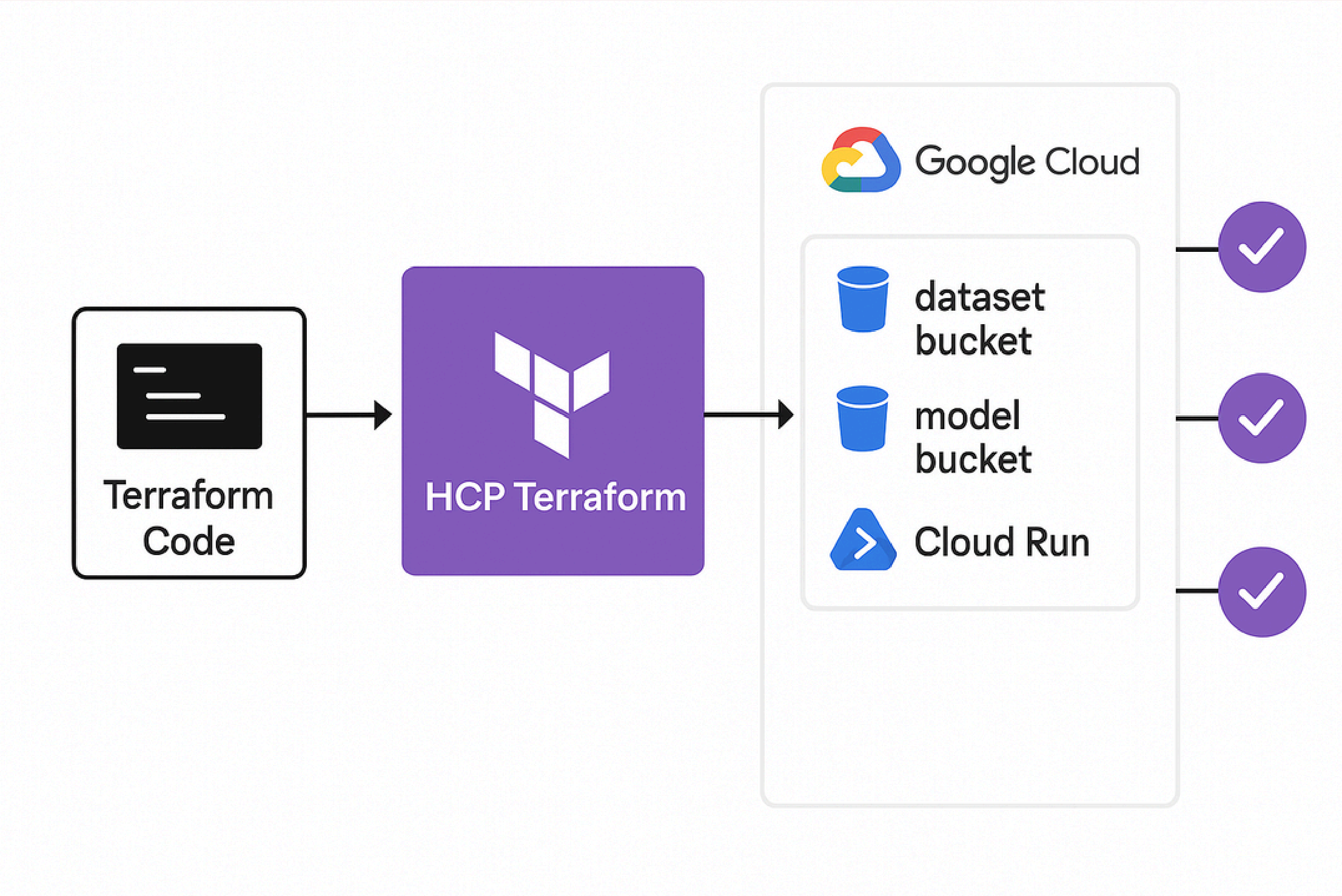


Terraform

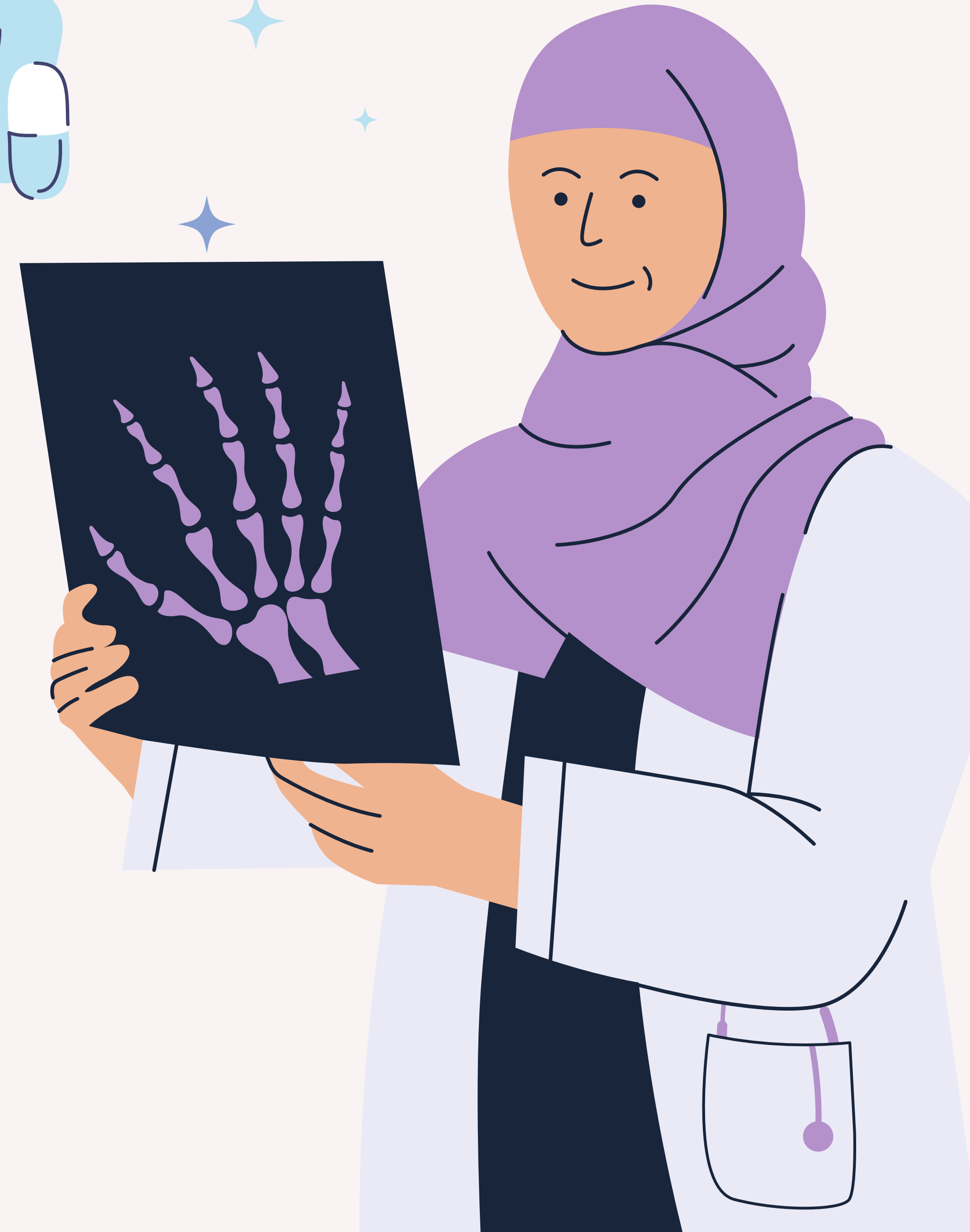
Terraform is an infrastructure as code tool that lets you build, change, and version infrastructure efficiently

As seen during the theoretical lesson multi-environment approach lets you build, test, and release code with greater speed and frequency to make your deployment as straightforward as possible.

The idea we had (even though we did not deploy it due to cost) was to leverage Github Secrets and Terraform to build a multitude of environment




An Eye on The Horizon

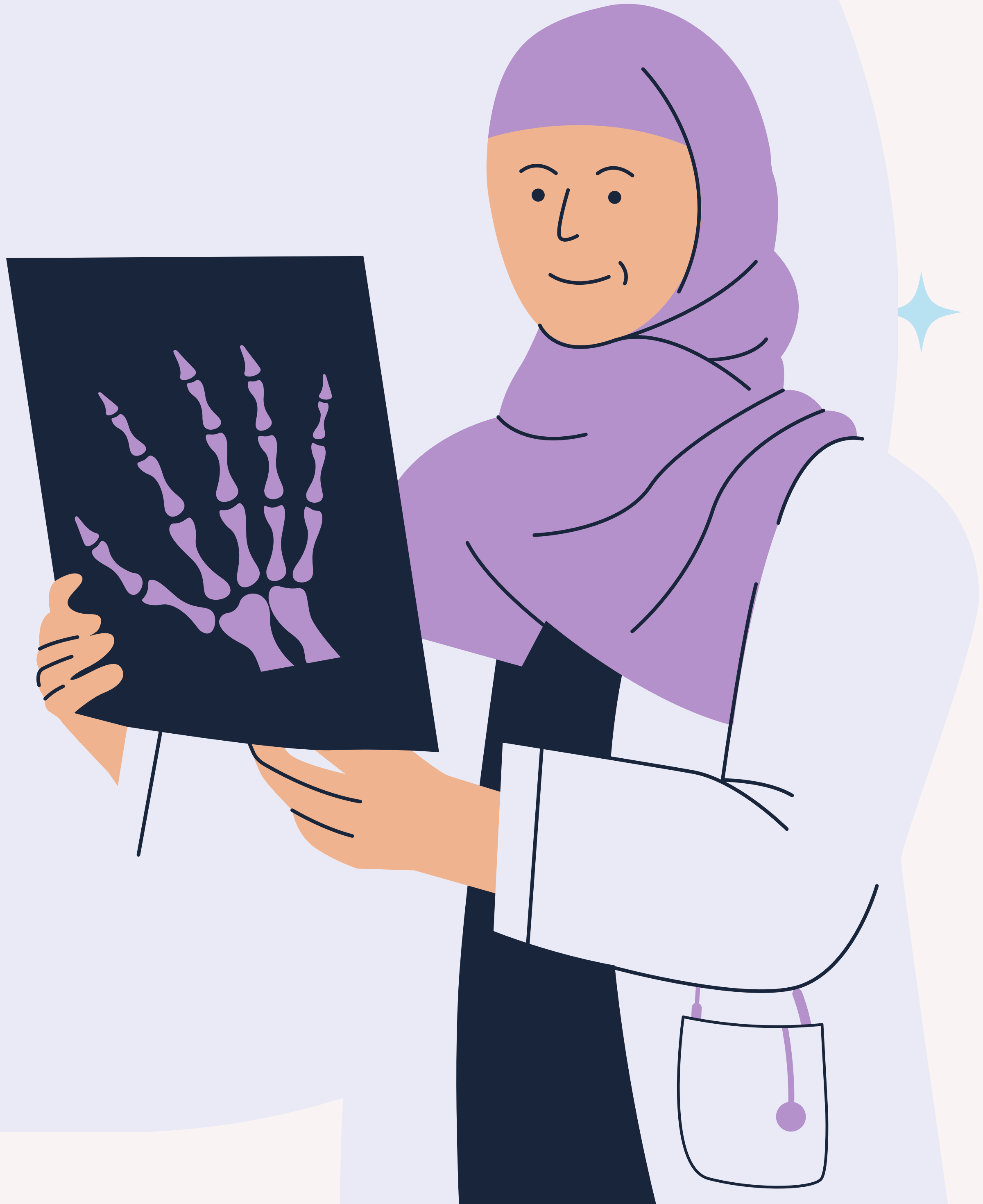
The Futur of Deepskin

An hypothetical futur.

Let's imagine for a moment that our application is successful. Suddenly, there's a need to scale this would require bringing in more developers, expanding to multiple regions, and supporting multi-environments.

Tools Required

As explained, Terraform would be a great tool to support our newly formed multi-environment solution. Jira, for example, could also be used to synchronize all the developers we now have.



Security and Costs

In this hypothetical future, as the application grows, we might need to expand our dataset, potentially using user feedback, to cover a wider range of skin conditions and retrain our entire model. Additionally, security would need to be strengthened, as medical data is highly sensitive and critical

The Futur of DEEPSKIN

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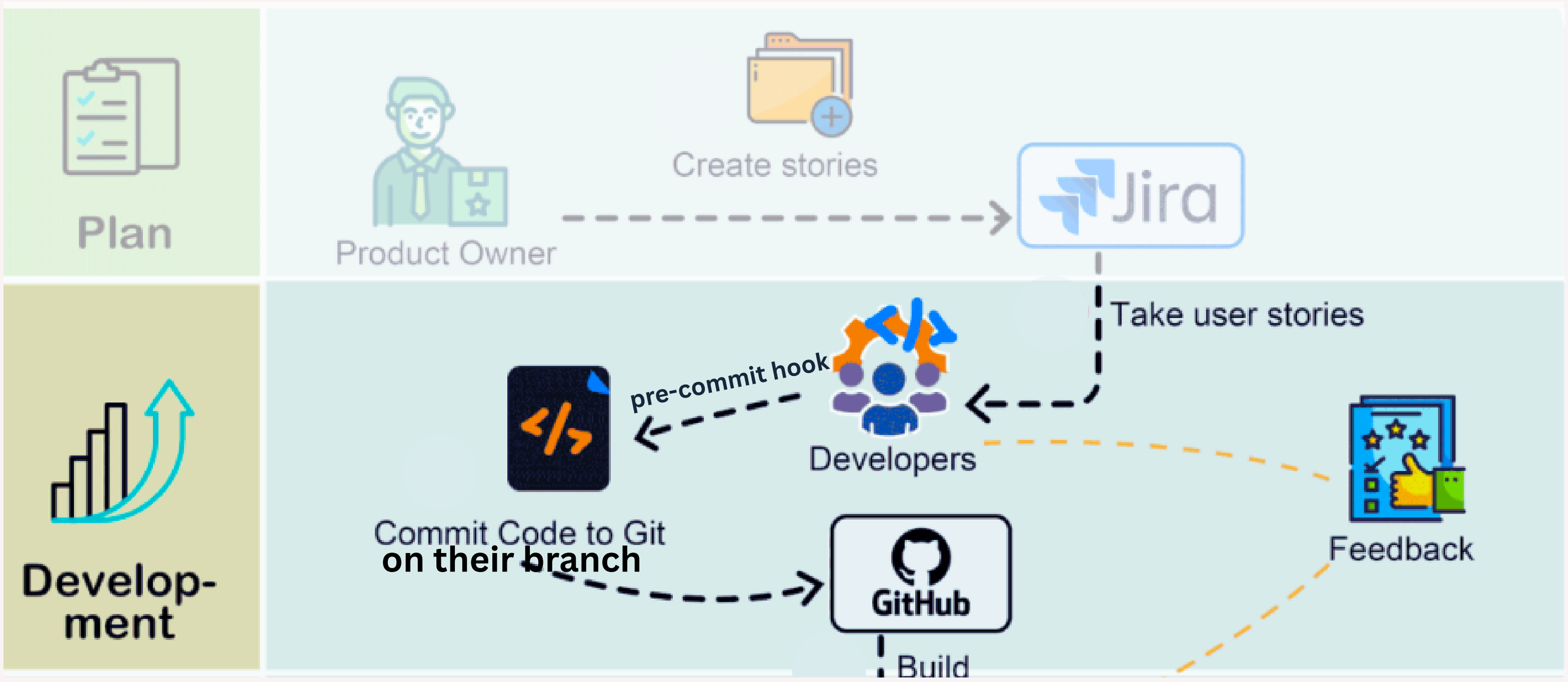
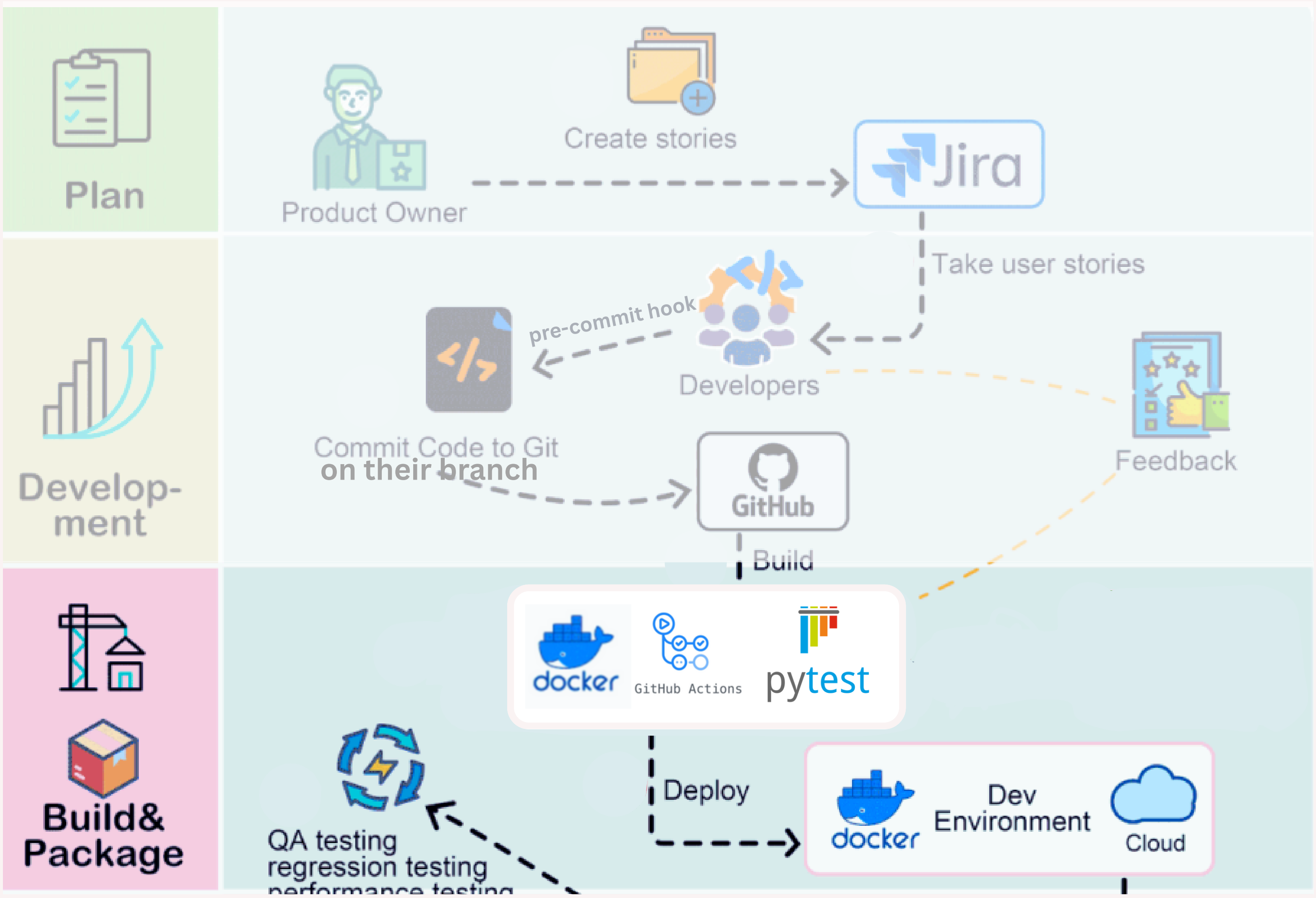


The Futur of DEEPSKIN



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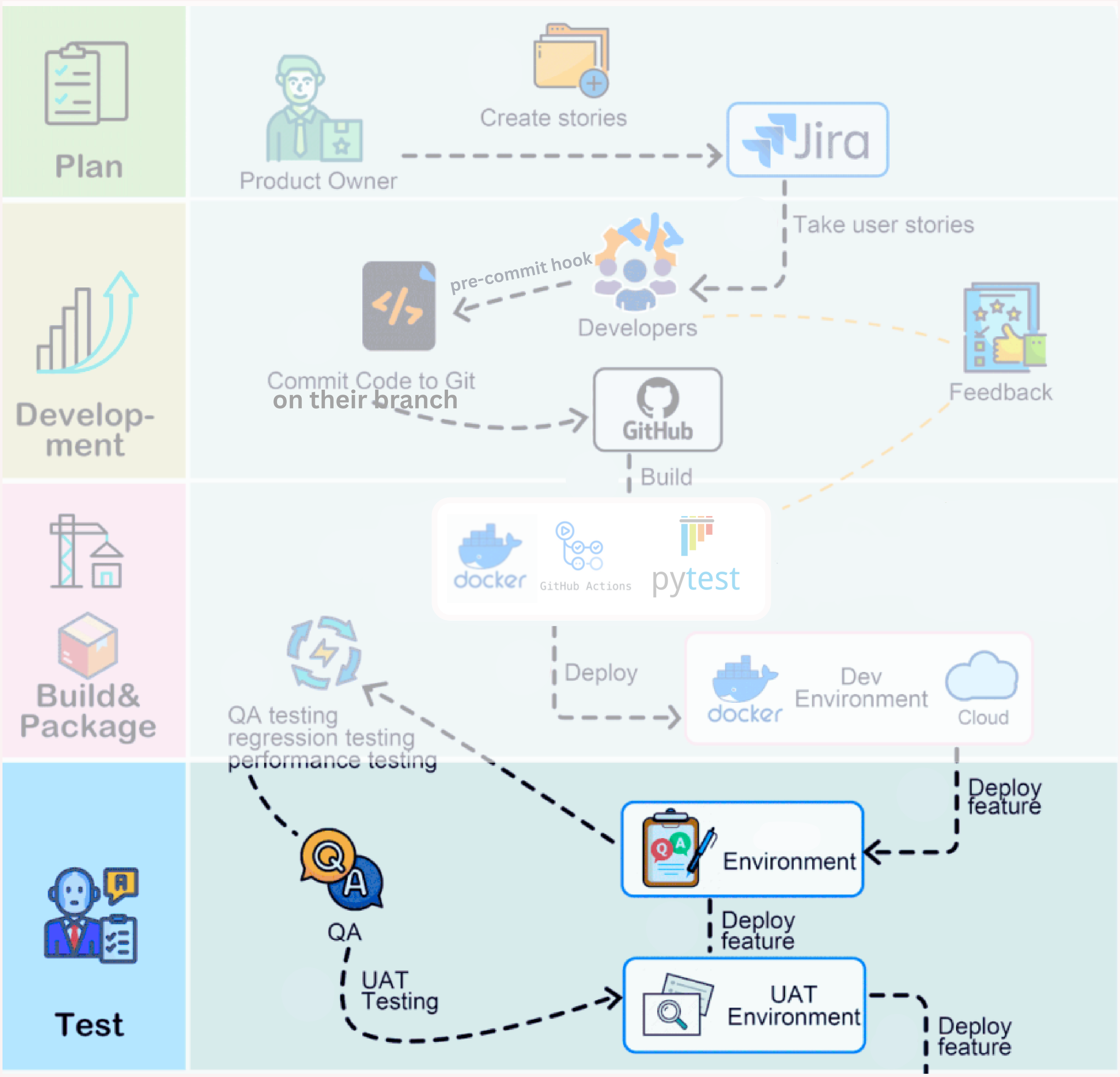

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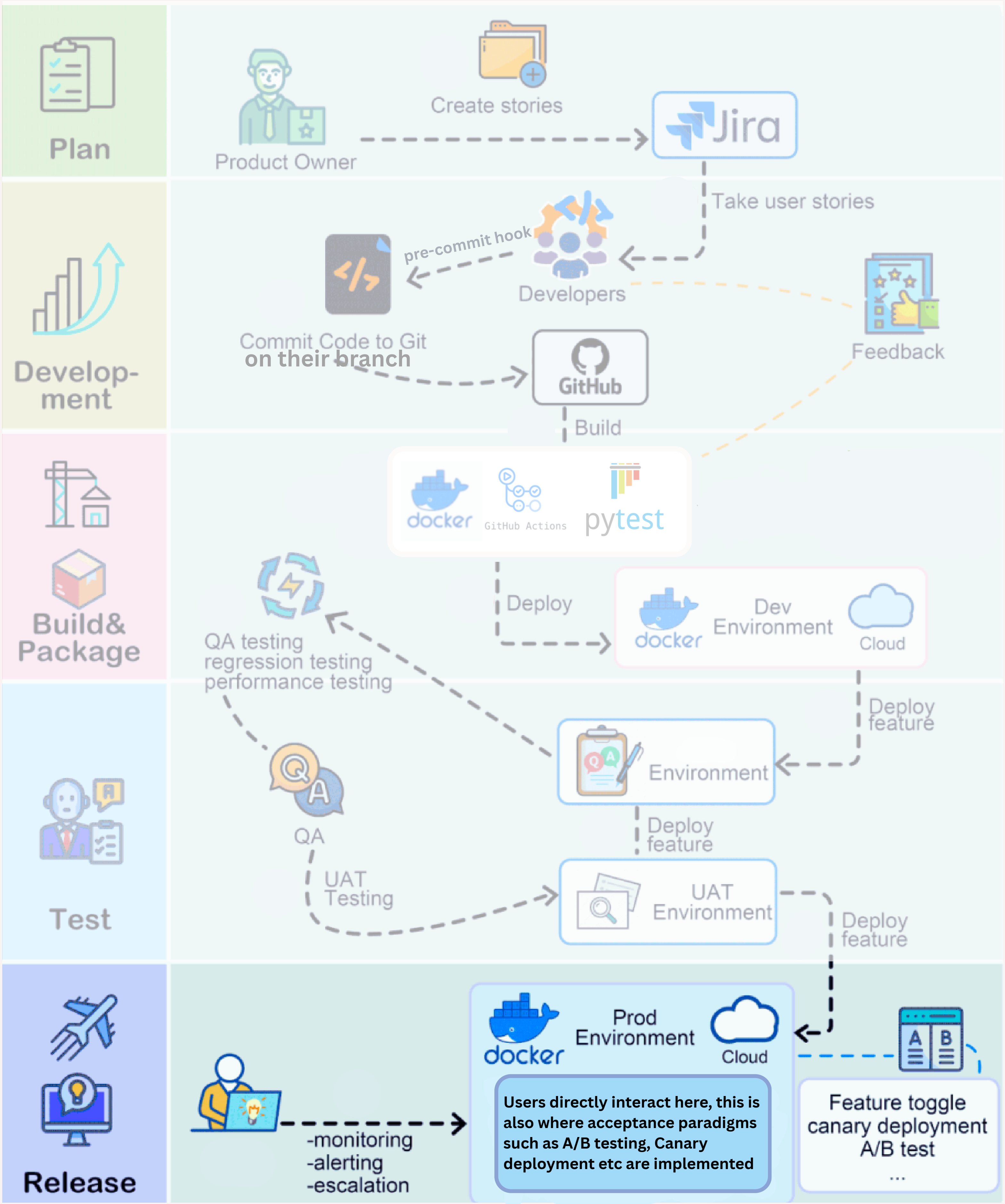
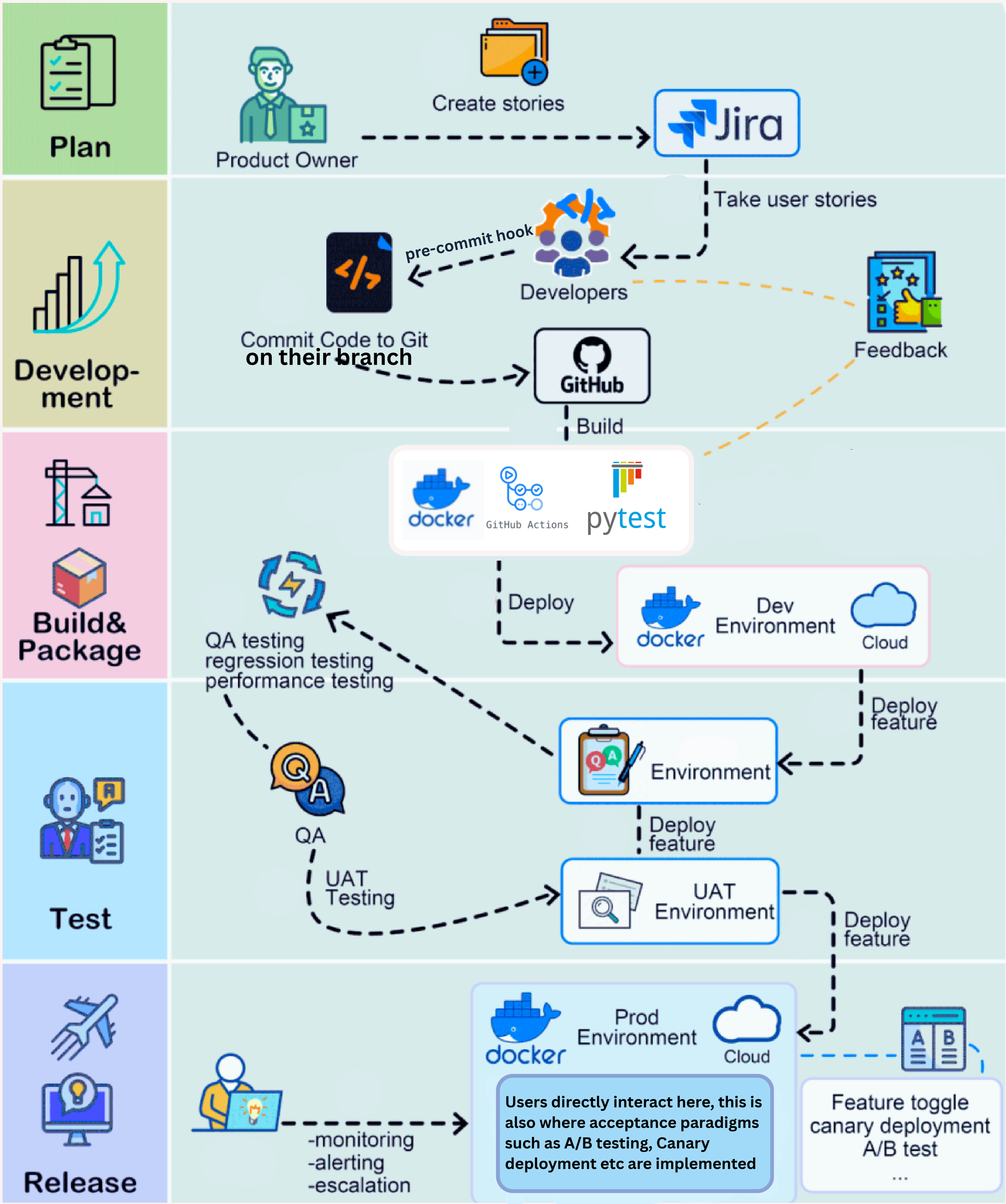


The Futur of DEEPSKIN



Thank you for your attention

Feel Free to ask any Questions